

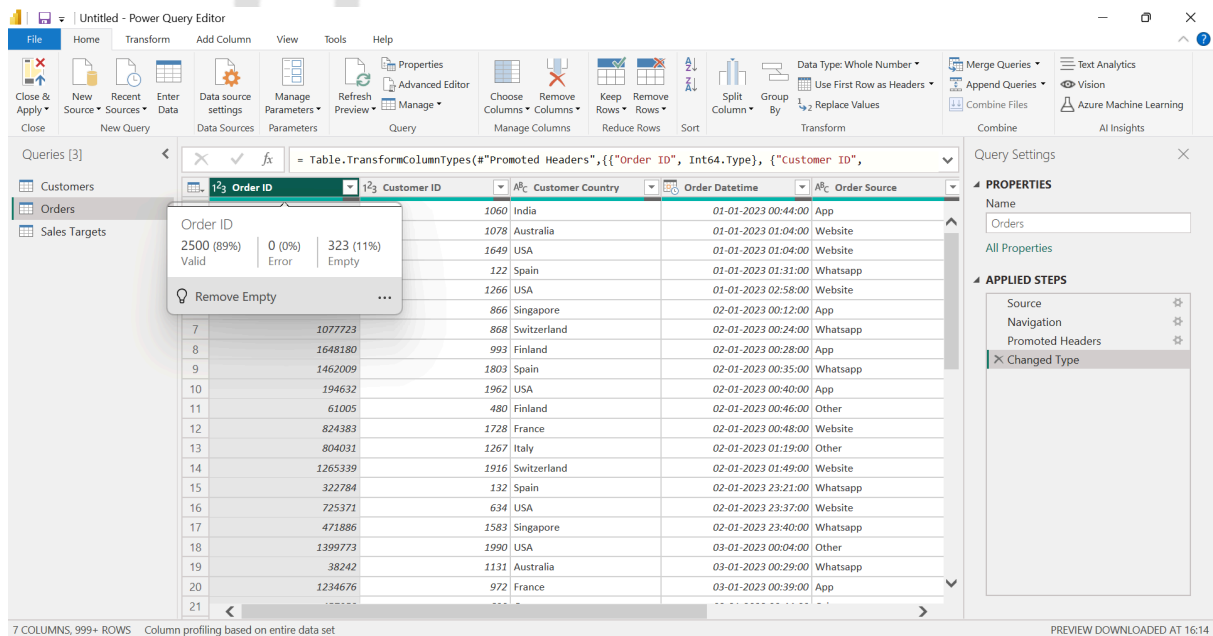
DIY PROJECT: E-COMMERCE ORDERS | SOLUTION

This document consists the stepwise solution for the Power BI Project and file below is the final PBIX Solution for the same:

[Link to Solution File](#)(Open in a new Tab)

Solutions

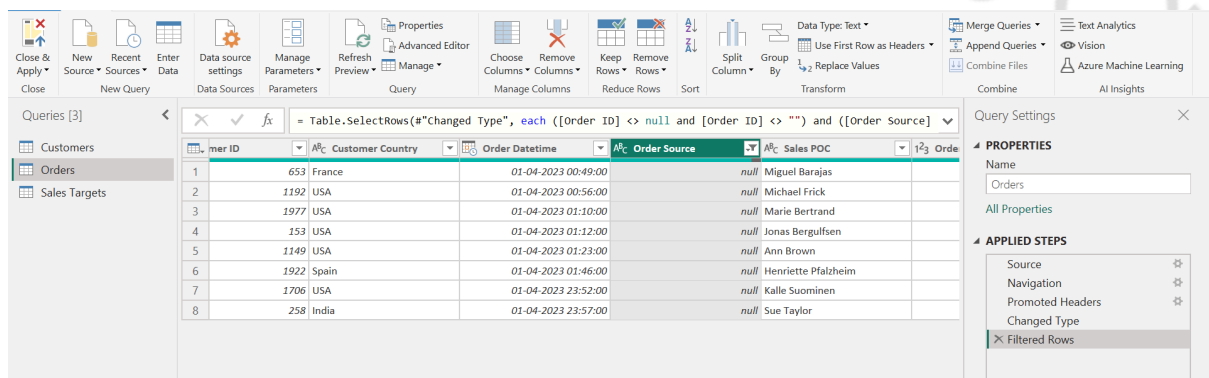
1. After importing data, load Power Query Editor
 - a. See if Power BI added extra nulls rows at the bottom of the dataset. If yes, remove them by using Remove Empty.



7 COLUMNS, 999+ ROWS Column profiling based on entire data set

PREVIEW DOWNLOADED AT 16:14

2. To see the missing order source (sales channel), use filter - filter order source for null values

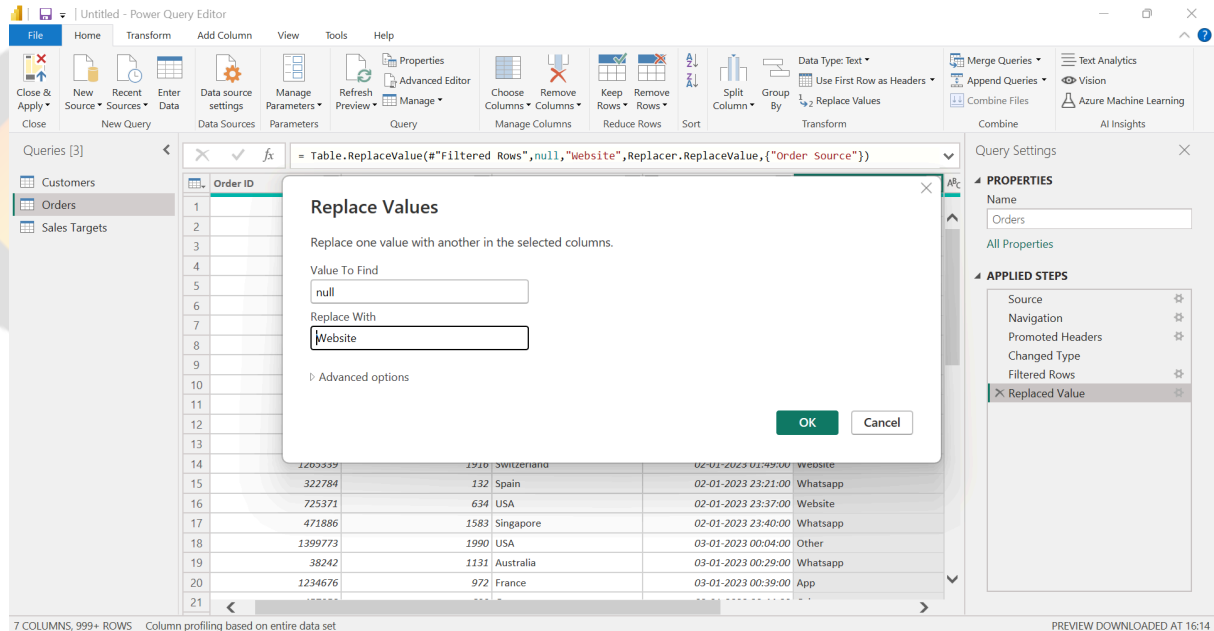


Query Settings

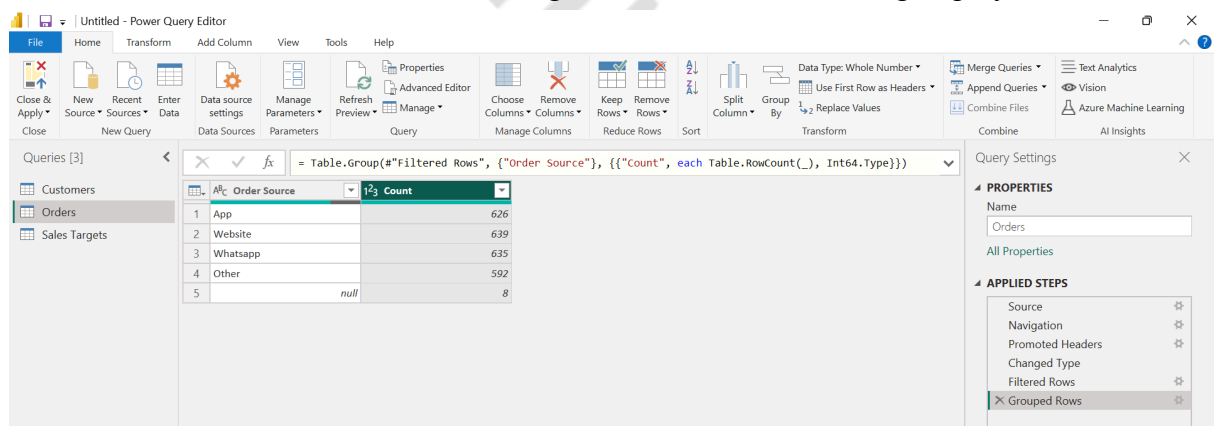
NAME: Orders

APPLIED STEPS: Source, Navigation, Promoted Headers, Changed Type, Filtered Rows

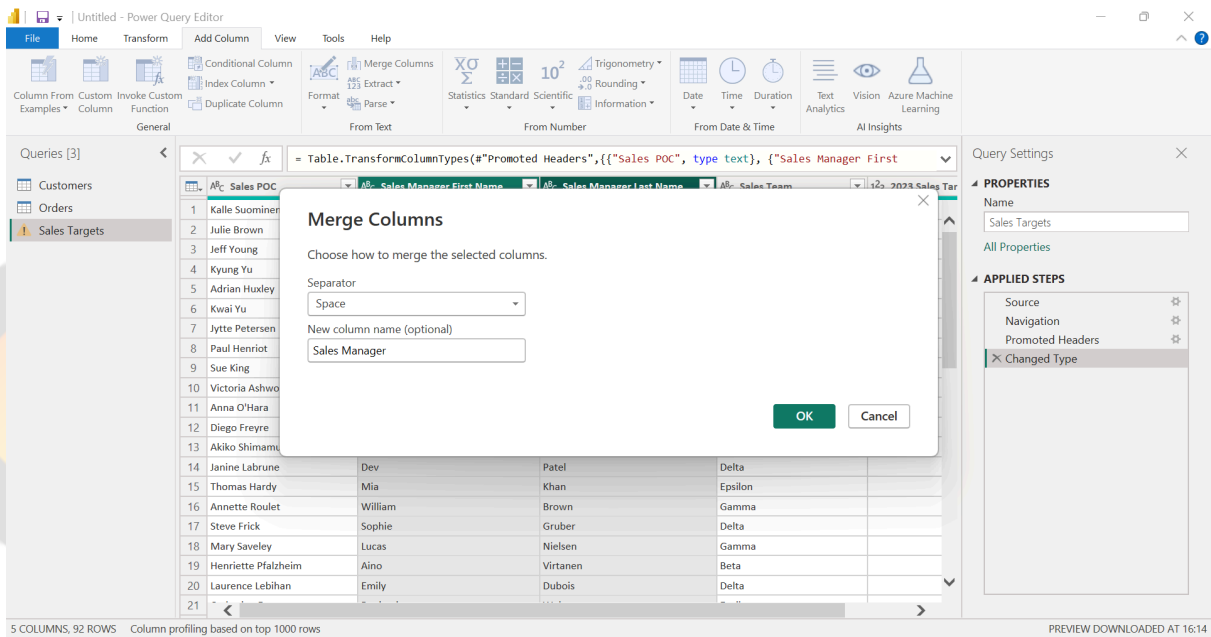
- The first and last timestamps when we weren't capturing this data can be seen from the image above.
- Replace these null values by the Order Source that occurs most frequently: i.e. Website.



- Note:** to find most occurring order source, we can use group by as shown here:

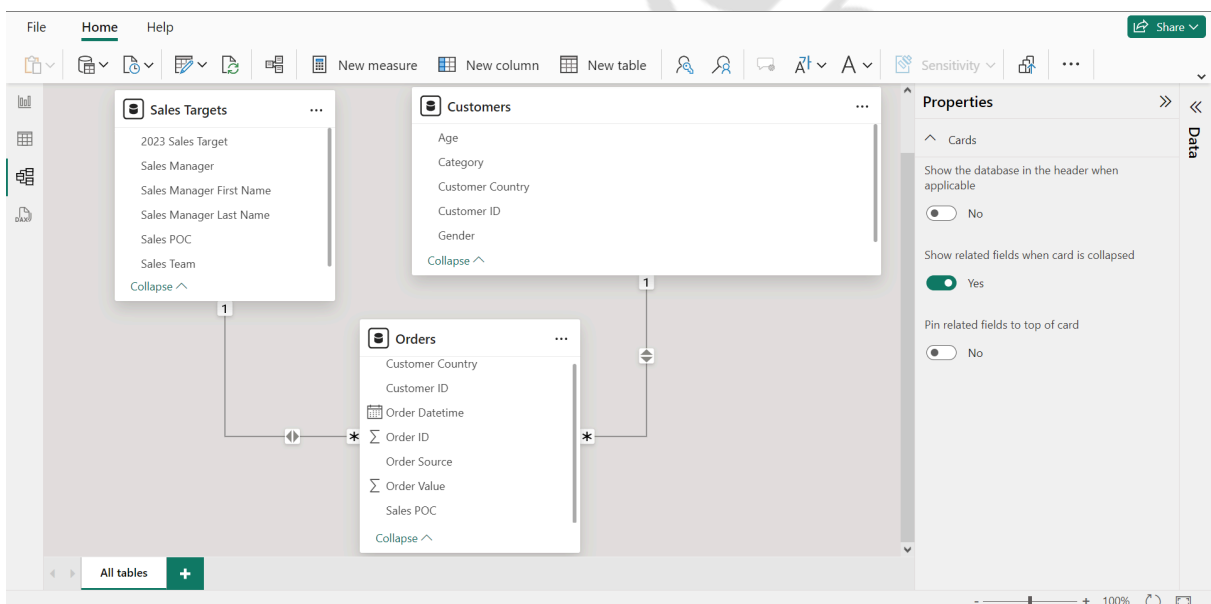


- Use Add Column → Merge Columns to create a Sales Manager Column



Load the data by clicking Close and Apply after this step.

4. Go to Model View and create relationships to ensure the following:
 - a. Orders table is related to Sales Targets by Sales POC in a 1:many bidirectional relationship (from Sales Targets to Orders)
 - b. Orders table is related to Customers by Sales POC in a 1:many single bidirectional relationship (from customers to Orders)
 - c. Bidirectional is needed here because we want the filters to work in both directions



5.

- a. Create a new column in the Sales Targets table that calculates Sales for each Sales POC

File Home Help Table tools Column tools

Name SalesbyPOC 123 Whole number \$% Whole number \$ % , .00 0 Σ Don't summarize

1 SalesbyPOC = SUMX(FILTER(Orders, Orders[Sales POC]='Sales Targets'[Sales POC]), Orders[Order Value])

Sales POC	Sales Manager First Name	Sales Manager Last Name	Sales Team	2023 Sales Target	Sales Manager	SalesbyPOC
Adrian Huxley	Chloe	Dupont	Beta	200000	Chloe Dupont	182094
Akiko Shimamura	Ava	Williams	Epsilon	100000	Ava Williams	88953
Allen Nelson	Alexander	Müller	Alpha	100000	Alexander Müller	116145
Ann Brown	Marco	Bianchi	Alpha	100000	Marco Bianchi	89991
Anna O'Hara	Isabella	Garcia	Alpha	200000	Isabella Garcia	242439
Annette Roulet	William	Brown	Gamma	100000	William Brown	84119
Arnold Cruz	Liam	Baumann	Beta	100000	Liam Baumann	132192
Carine Schmitt	Aino	Virtanen	Beta	30000	Aino Virtanen	38017
Catherine Dewey	Benjamin	Weber	Epsilon	100000	Benjamin Weber	98950
Christina Berqlund	Olivia	Jensen	Beta	100000	Olivia Jensen	110015
Dan Lewis	Emily	Dubois	Delta	30000	Emily Dubois	42553
Daniel Da Cunha	Lucas	Olsen	Gamma	100000	Lucas Olsen	108833
Daniel Tonini	William	Brown	Gamma	100000	William Brown	86855
Dean Cassidy	Oliver	Kumar	Beta	50000	Oliver Kumar	57084
Diego Freyre	Noah	Meier	Alpha	1000000	Noah Meier	1089868
Dominique Perrier	Liam	O'Sullivan	Beta	150000	Liam O'Sullivan	143869
Eduardo Saavedra	Emily	Murphy	Alpha	50000	Emily Murphy	45184
Elizabeth Devon	Dev	Patel	Delta	100000	Dev Patel	107329
Elizabeth Lincoln	Alexander	Müller	Alpha	80000	Alexander Müller	71209
Eric Natividad	Daniela	Hernandez	Gamma	200000	Daniela Hernandez	201398
Francisca Cervantes	Dev	Patel	Delta	100000	Dev Patel	88589
Frederique Citeaux	Daniela	Hernandez	Gamma	100000	Daniela Hernandez	94689
Georg Pippis	Priya	Kapoor	Alpha	150000	Priya Kapoor	188709

Table: Sales Targets (92 rows) Column: SalesbyPOC (92 distinct values)

Data

- Customers
- Orders
 - Customer Country
 - Customer ID
 - Order Datetime
 - Order ID
 - Order Source
 - Order Value
 - Sales POC
- Sales Targets
 - 2023 Sales Target
 - Sales Manager
 - Sales Manager First Name
 - Sales Manager Last Name
 - Sales POC
 - Sales Team

SalesbyPOC = SUMX(FILTER(Orders, Orders[Sales POC]='Sales Targets'[Sales POC]), Orders[Order Value])

File Home Help Table tools Column tools

Name SalesbyPOC 123 Whole number \$% Whole number \$ % , .00 0 Σ Don't summarize

1 SalesbyPOC = SUMX(FILTER(Orders, Orders[Sales POC]='Sales Targets'[Sales POC]), Orders[Order Value])

Sales POC	Sales Manager First Name	Sales Manager Last Name	Sales Team	2023 Sales Target	Sales Manager	SalesbyPOC
Adrian Huxley	Chloe	Dupont	Beta	200000	Chloe Dupont	182094
Akiko Shimamura	Ava	Williams	Epsilon	100000	Ava Williams	88953
Allen Nelson	Alexander	Müller	Alpha	100000	Alexander Müller	116145
Ann Brown	Marco	Bianchi	Alpha	100000	Marco Bianchi	89991
Anna O'Hara	Isabella	Garcia	Alpha	200000	Isabella Garcia	242439
Annette Roulet	William	Brown	Gamma	100000	William Brown	84119
Arnold Cruz	Liam	Baumann	Beta	100000	Liam Baumann	132192
Carine Schmitt	Aino	Virtanen	Beta	30000	Aino Virtanen	38017
Catherine Dewey	Benjamin	Weber	Epsilon	100000	Benjamin Weber	98950
Christina Berqlund	Olivia	Jensen	Beta	100000	Olivia Jensen	110015
Dan Lewis	Emily	Dubois	Delta	30000	Emily Dubois	42553
Daniel Da Cunha	Lucas	Olsen	Gamma	100000	Lucas Olsen	108833
Daniel Tonini	William	Brown	Gamma	100000	William Brown	86855
Dean Cassidy	Oliver	Kumar	Beta	50000	Oliver Kumar	57084
Diego Freyre	Noah	Meier	Alpha	1000000	Noah Meier	1089868
Dominique Perrier	Liam	O'Sullivan	Beta	150000	Liam O'Sullivan	143869
Eduardo Saavedra	Emily	Murphy	Alpha	50000	Emily Murphy	45184
Elizabeth Devon	Dev	Patel	Delta	100000	Dev Patel	107329
Elizabeth Lincoln	Alexander	Müller	Alpha	80000	Alexander Müller	71209
Eric Natividad	Daniela	Hernandez	Gamma	200000	Daniela Hernandez	201398
Francisca Cervantes	Dev	Patel	Delta	100000	Dev Patel	88589
Frederique Citeaux	Daniela	Hernandez	Gamma	100000	Daniela Hernandez	94689
Georg Pippis	Priya	Kapoor	Alpha	150000	Priya Kapoor	188709

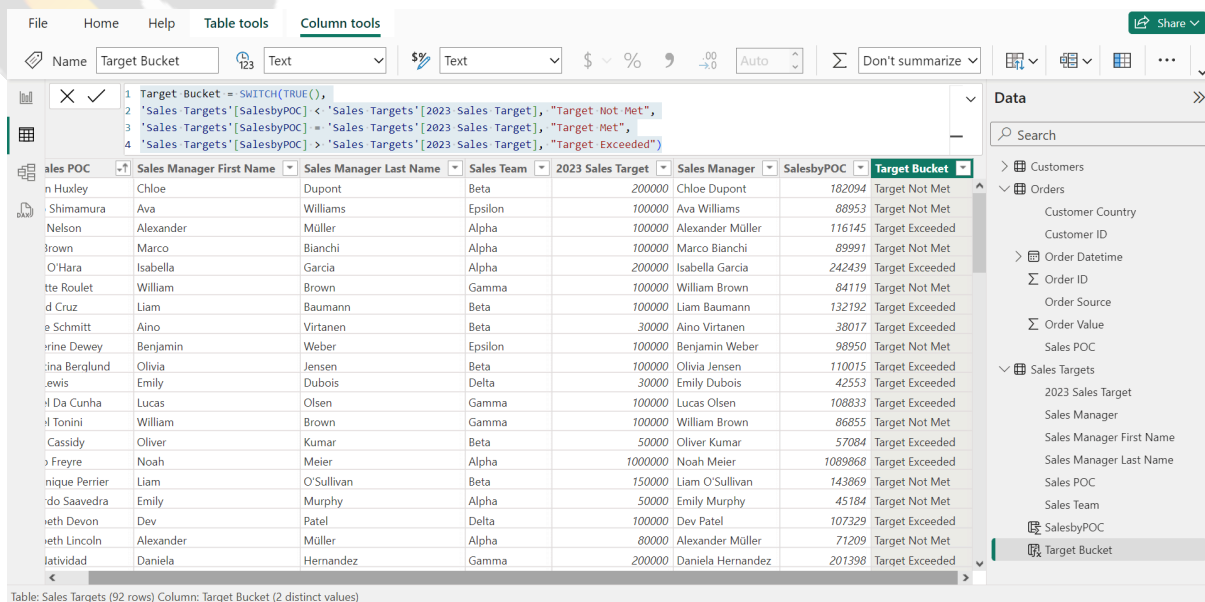
Table: Sales Targets (92 rows) Column: SalesbyPOC (92 distinct values)

Data

- Customers
- Orders
 - Customer Country
 - Customer ID
 - Order Datetime
 - Order ID
 - Order Source
 - Order Value
 - Sales POC
- Sales Targets
 - 2023 Sales Target
 - Sales Manager
 - Sales Manager First Name
 - Sales Manager Last Name
 - Sales POC
 - Sales Team

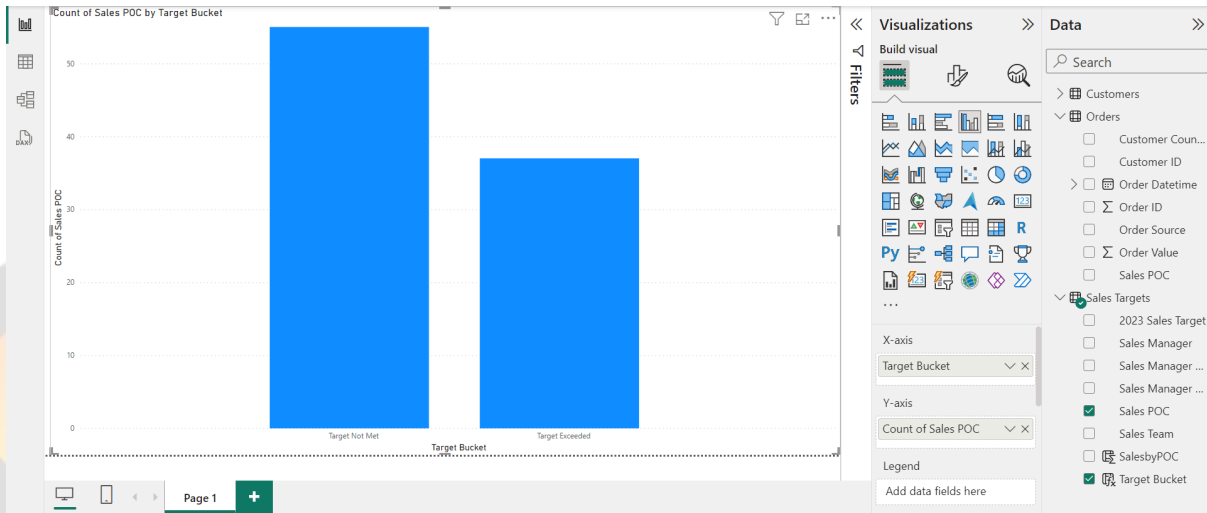
Use DAX SWITCH function to define three buckets.
This formula can be used:

Target Bucket = SWITCH(TRUE(),
'Sales Targets'[SalesbyPOC] < 'Sales Targets'[2023 Sales Target], "Target Not
Met",
'Sales Targets'[SalesbyPOC] = 'Sales Targets'[2023 Sales Target], "Target
Met",
'Sales Targets'[SalesbyPOC] > 'Sales Targets'[2023 Sales Target], "Target
Exceeded")

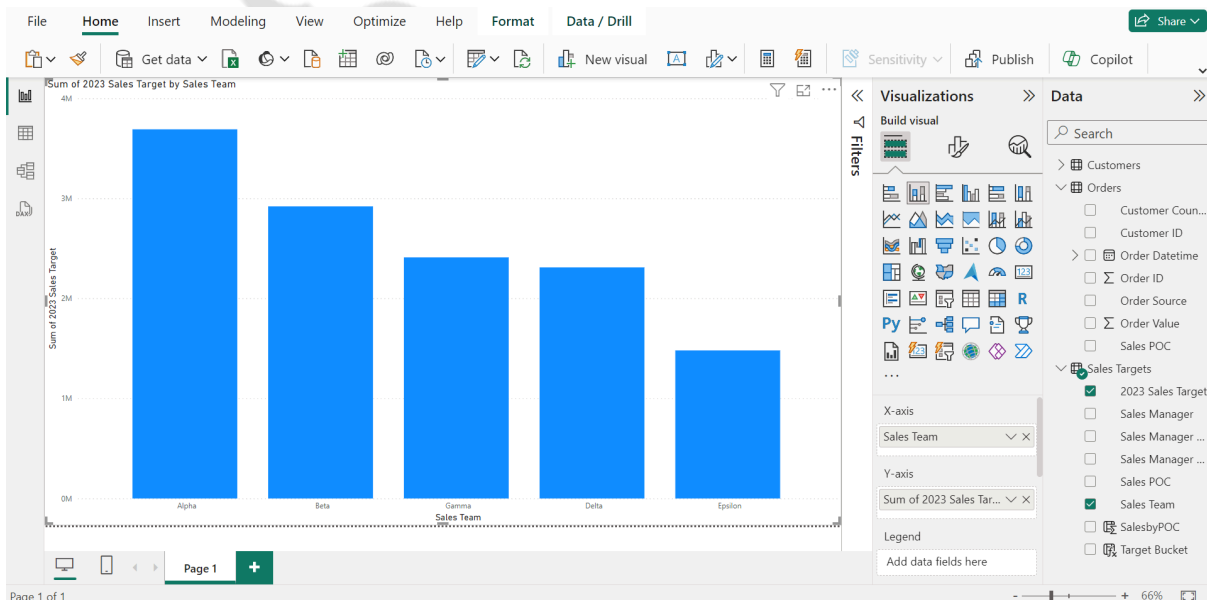


Sales Manager First Name	Sales Manager Last Name	Sales Team	2023 Sales Target	Sales Manager	SalesbyPOC	Target Bucket
Chloe Dupont	Dupont	Beta	200000	Chloe Dupont	182094	Target Not Met
Ava Williams	Williams	Epsilon	100000	Ava Williams	88953	Target Not Met
Alexander Müller	Müller	Alpha	100000	Alexander Müller	116145	Target Exceeded
Marco Bianchi	Bianchi	Alpha	100000	Marco Bianchi	89991	Target Not Met
Isabella Garcia	Garcia	Alpha	200000	Isabella Garcia	242439	Target Exceeded
William Brown	Brown	Gamma	100000	William Brown	84119	Target Not Met
Liam Baumann	Baumann	Beta	100000	Liam Baumann	132192	Target Exceeded
Aino Virtanen	Virtanen	Beta	30000	Aino Virtanen	38017	Target Exceeded
Benjamin Weber	Weber	Epsilon	100000	Benjamin Weber	98950	Target Not Met
Olivia Jensen	Jensen	Beta	100000	Olivia Jensen	110015	Target Exceeded
Emily Dubois	Dubois	Delta	30000	Emily Dubois	42553	Target Exceeded
Lucas Olsen	Olsen	Gamma	100000	Lucas Olsen	108833	Target Exceeded
William Brown	Brown	Gamma	100000	William Brown	86855	Target Not Met
Oliver Kumar	Kumar	Beta	50000	Oliver Kumar	57084	Target Exceeded
Noah Meier	Meier	Alpha	1000000	Noah Meier	1089868	Target Exceeded
Liam O'Sullivan	O'Sullivan	Beta	150000	Liam O'Sullivan	143869	Target Not Met
Emily Murphy	Murphy	Alpha	50000	Emily Murphy	45184	Target Not Met
Dev Patel	Patel	Delta	100000	Dev Patel	107329	Target Exceeded
Alexander Müller	Müller	Alpha	80000	Alexander Müller	71209	Target Not Met
Daniela Hernandez	Hernandez	Gamma	200000	Daniela Hernandez	201398	Target Exceeded

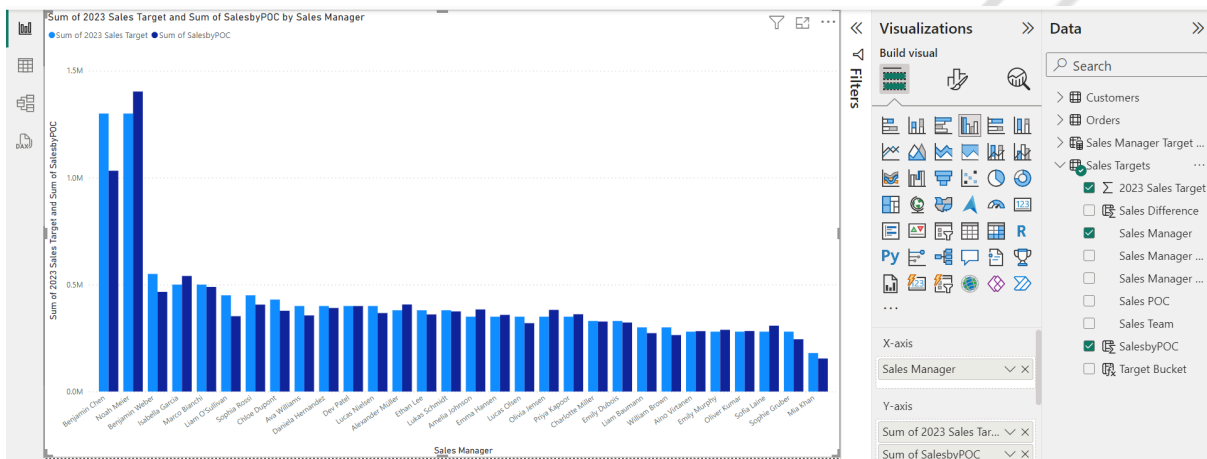
Create a chart to show these categories:



Create a chart which has bars/columns for each Sales Team:

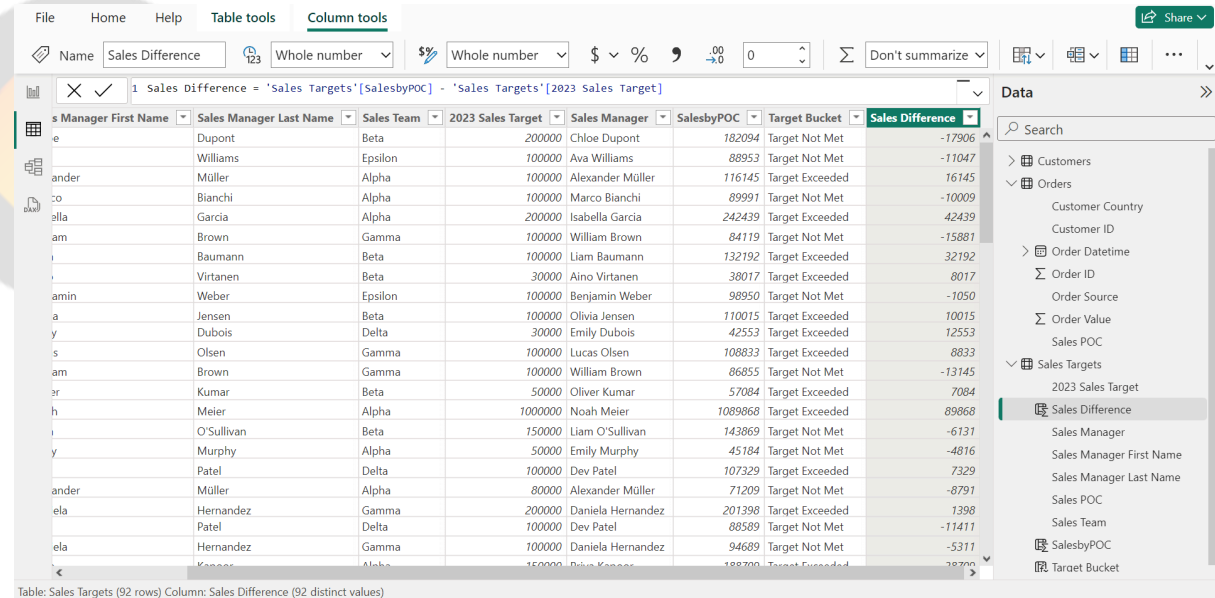


Create a chart for total target for each Sales Manager:



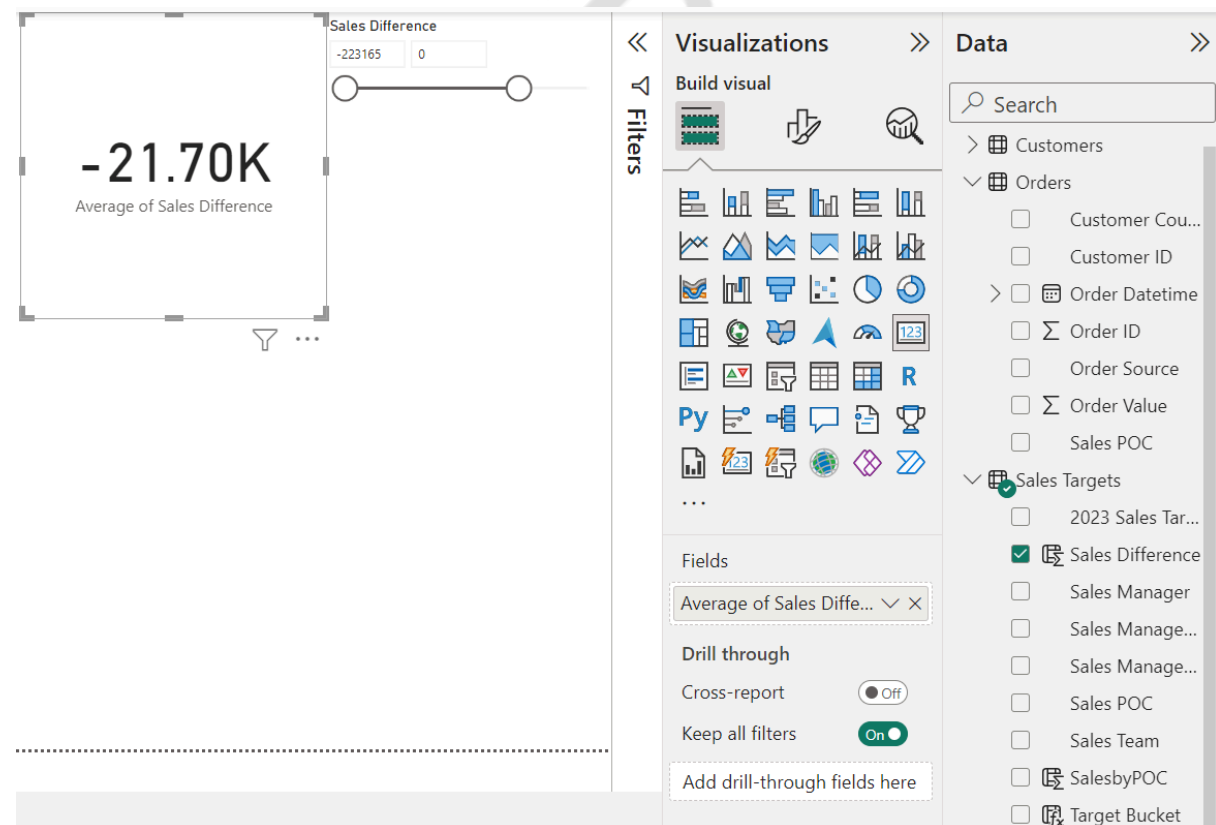
Benjamin Chen and Noah Meier were given the highest target. Benjamin Chen did not meet the sales target while Noah Meier met it.

- Create a new DAX column for finding the difference between the Actual Sales and the Target. If it is negative, it means that target has not been met.



Manager First Name	Sales Manager Last Name	Sales Team	2023 Sales Target	Sales Manager	SalesbyPOC	Target Bucket	Sales Difference
e	Dupont	Beta	200000	Chloe Dupont	182094	Target Not Met	-17906
	Williams	Epsilon	100000	Ava Williams	88953	Target Not Met	-11047
ander	Müller	Alpha	100000	Alexander Müller	116145	Target Exceeded	16145
co	Bianchi	Alpha	100000	Marco Bianchi	89991	Target Not Met	-10009
ella	Garcia	Alpha	200000	Isabella Garcia	242439	Target Exceeded	42439
am	Brown	Gamma	100000	William Brown	84119	Target Not Met	-15881
y	Baumann	Beta	100000	Liam Baumann	132192	Target Exceeded	32192
	Virtanen	Beta	30000	Aino Virtanen	38017	Target Exceeded	8017
amin	Weber	Epsilon	100000	Benjamin Weber	98950	Target Not Met	-1050
a	Jensen	Beta	100000	Olivia Jensen	110015	Target Exceeded	10015
y	Dubois	Delta	30000	Emily Dubois	42553	Target Exceeded	12553
s	Olsen	Gamma	100000	Lucas Olsen	108833	Target Exceeded	8833
am	Brown	Gamma	100000	William Brown	86855	Target Not Met	-13145
er	Kumar	Beta	50000	Oliver Kumar	57084	Target Exceeded	7084
h	Meier	Alpha	1000000	Noah Meier	1089868	Target Exceeded	89868
i	O'Sullivan	Beta	150000	Liam O'Sullivan	143869	Target Not Met	-6131
y	Murphy	Alpha	50000	Emily Murphy	45184	Target Not Met	-4816
	Patel	Delta	100000	Dev Patel	107329	Target Exceeded	7329
ander	Müller	Alpha	80000	Alexander Müller	71209	Target Not Met	-8791
ela	Hernandez	Gamma	200000	Daniela Hernandez	201398	Target Exceeded	1398
	Patel	Delta	100000	Dev Patel	88589	Target Not Met	-11411
ela	Hernandez	Gamma	100000	Daniela Hernandez	94689	Target Not Met	-5311

Now we need to simply filter this column for negative values and find the average. We will do this using a slicer and a Card visual for the average.



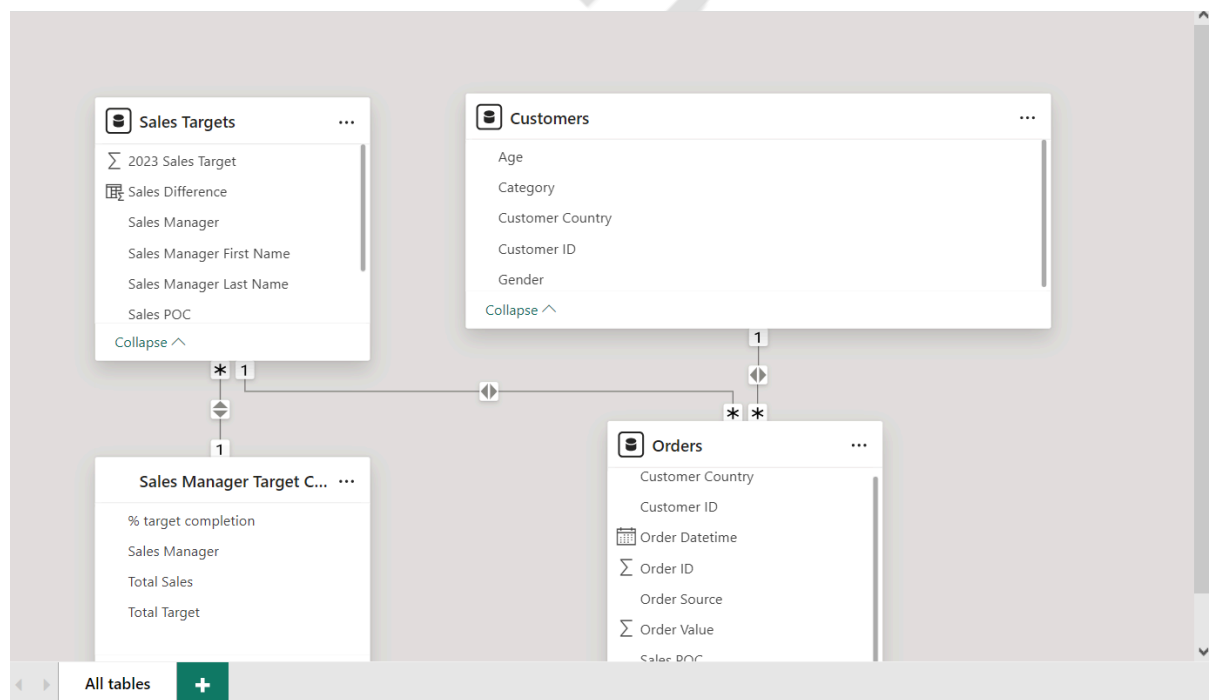
Create a new Sales Manager Targets table as shown below

1 Sales Manager Target Completion = SUMMARIZE('Sales Targets', 'Sales Targets'[Sales Manager], "Total Sales", SUM('Sales Targets'[SalesbyPOC]), "Total Target", SUM('Sales Targets'[2023 Sales Target]), "% target completion", 100*(SUM('Sales Targets'[SalesbyPOC])/SUM('Sales Targets'[2023 Sales Target])))

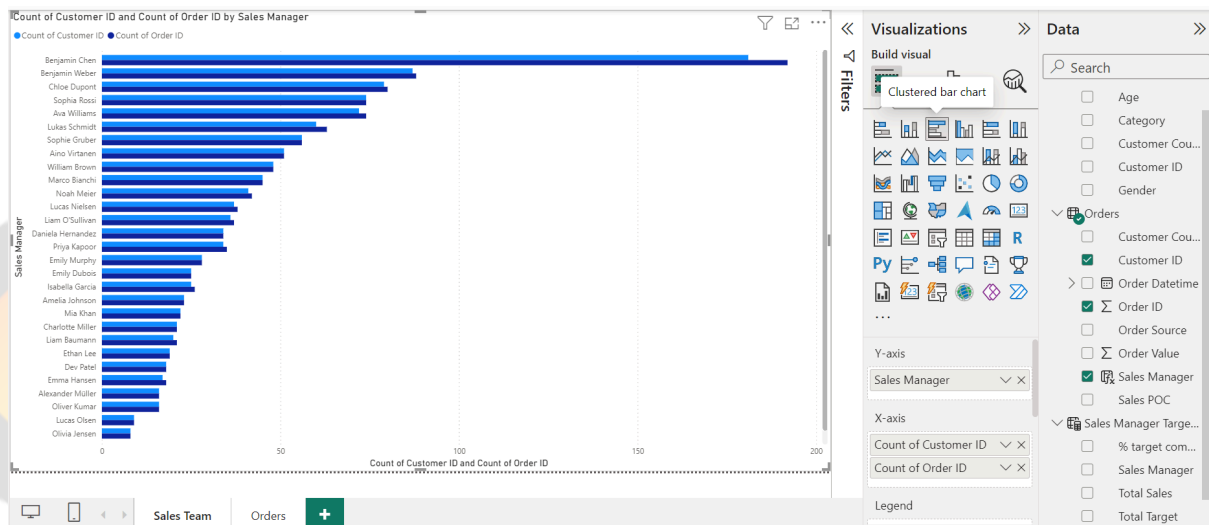
Sales Manager	% target completion	Total Sales	Total Target
Amelia Johnson	109.641142857143	383744	350000
Lukas Schmidt	98.58	374604	380000
Olivia Jensen	109.085714285714	381800	350000
Sofia Laine	109.96	307888	280000
Chloe Dupont	87.8355813953488	377693	430000
Alexander Müller	107.128684210526	407089	380000
Liam O'Sullivan	78.268	352206	450000
Sophia Rossi	90.3317777777778	406493	450000
Emma Hansen	102.409714285714	358434	350000
Ethan Lee	94.8718421052632	360513	380000
Isabella Garcia	108.082	540410	500000
Noah Meier	107.918538461538	1402941	1300000
Ava Williams	88.97125	355885	400000
Dev Patel	99.95125	399805	400000
Mia Khan	85.7955555555556	154432	180000
William Brown	88.0126666666667	264038	300000
Sophie Gruber	87.2842857142857	244396	280000
Lucas Nielsen	91.74825	366993	400000
Aino Virtanen	100.861071428571	282411	280000
Emily Dubois	97.7675757575758	322633	330000
Benjamin Weber	84.7278181818182	466003	550000
Emily Murphy	103.081071428571	288627	280000

Table: Sales Manager Target Completion (30 rows) Column: Sales Manager (30 distinct values)

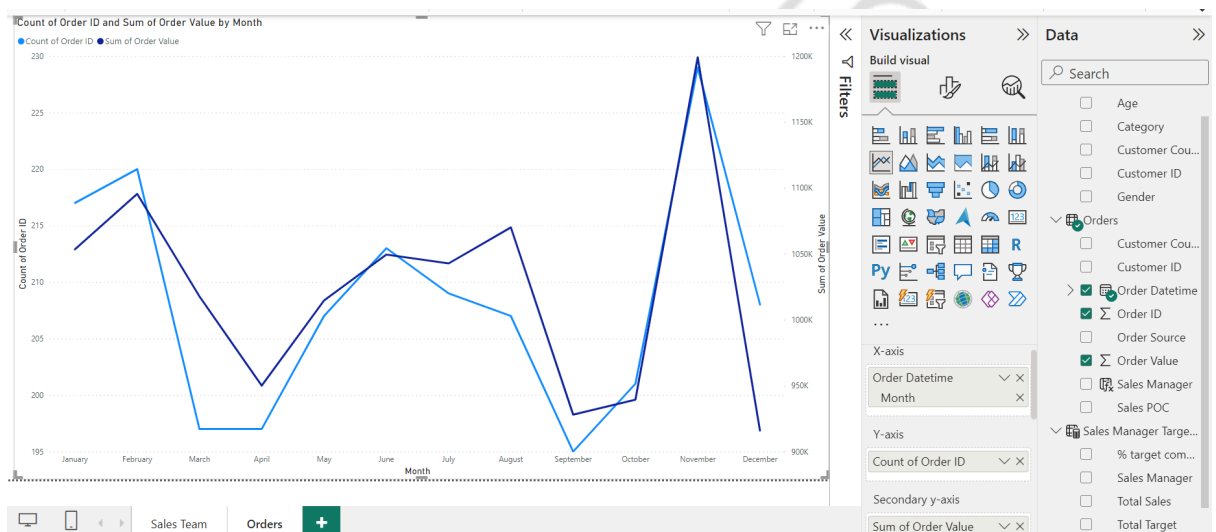
Add Sales Manager Target Completion Table to the Data Model so that filters work properly. A relationship with Sales Targets table can be created (1: many, bidirectional). This is shown below.



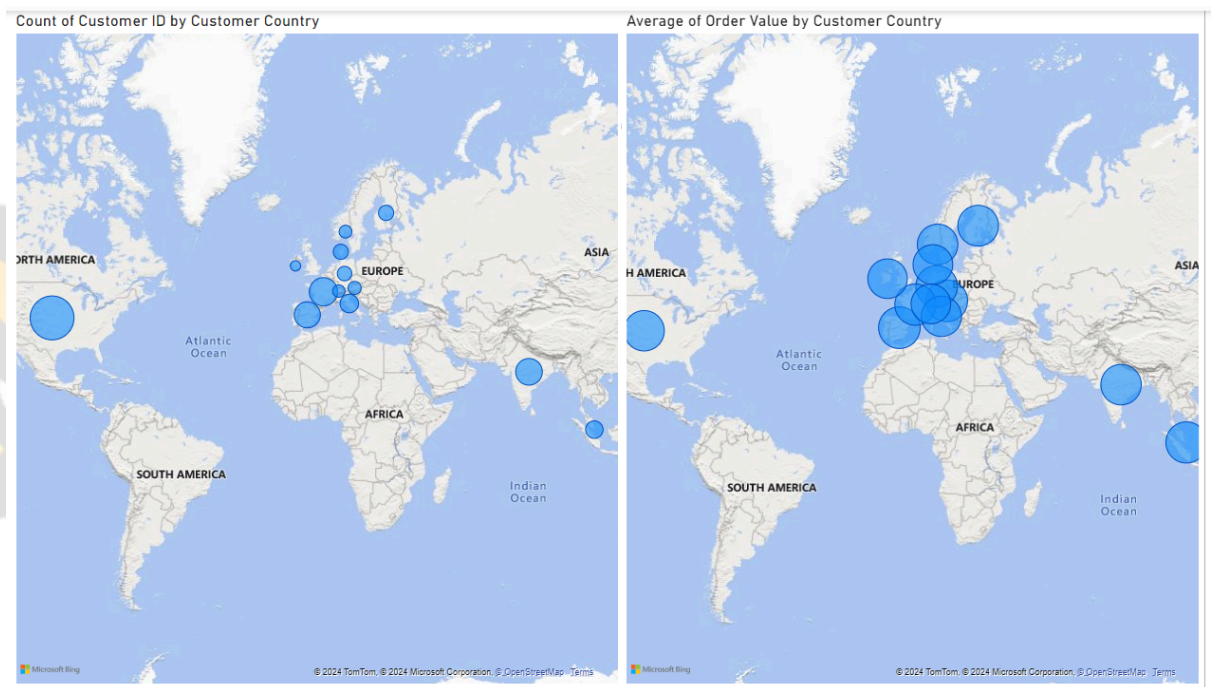
Create a chart to plot distinct count of Customer ID and order ID by Sales Manager



7. Create a line chart for demonstrating this trend



8. Plot the number of orders and average order value using map visualisations.



9. This one is slightly tricky and requires you to create a measure.
- First, create a calculated column: DidNotOrder which is 1 if there is an order corresponding to a customer ID, else 0.

DidNotOrder = IF(COUNTROWS(RELATEDTABLE(Orders)) = 0, 1, 0)

- Then create a measure to calculate the % of customers who did not order.

% Customers Did Not Order =

DIVIDE(
CALCULATE(SUM(Customers[DidNotOrder])),
CALCULATE(COUNTROWS(Customers)),
0
)

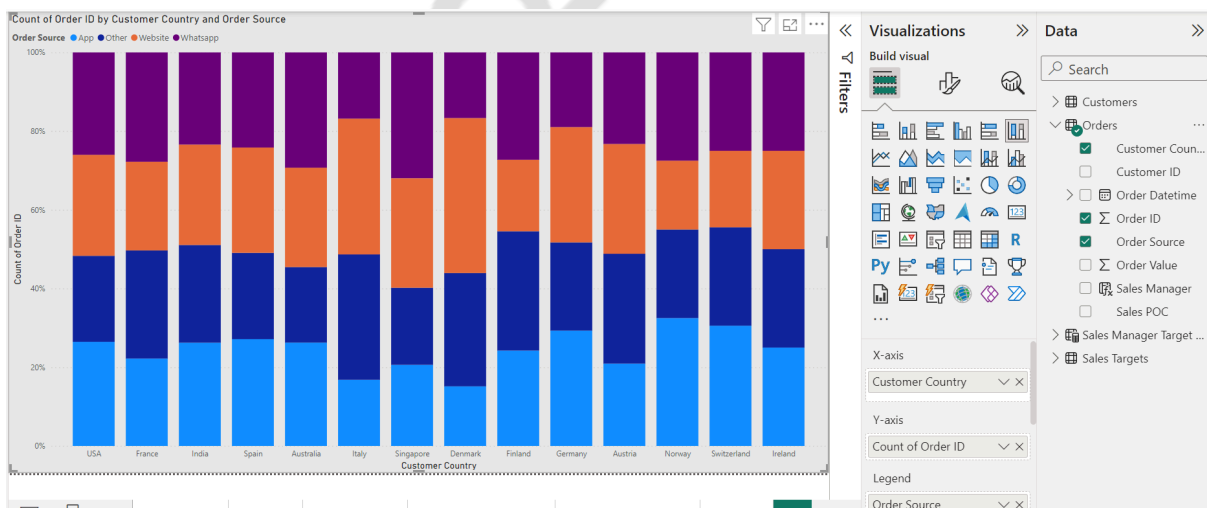
1	% Customers Did Not Order =
2	DIVIDE(
3	CALCULATE(SUM(Customers[DidNotOrder])),
4	CALCULATE(COUNTROWS(Customers)),
5	0
6	)

Customer ID	Customer Country	Gender	Age	Category	DidNotOrder
2003	USA	M	23	E	1
2006	USA	M	21	E	1
2011	USA	M	18	E	1
2012	USA	M	42	E	1
2013	USA	M	39	E	1
2017	USA	M	76	E	1
2019	USA	M	57	E	1
2023	USA	M	73	E	1
2029	USA	M	48	E	1
2033	USA	M	21	E	1
2036	USA	M	83	E	1
2048	USA	M	63	E	1
2049	USA	M	67	E	1
2053	USA	M	38	E	1
2056	USA	M	79	E	1
2058	USA	M	30	E	1
2061	USA	M	78	E	1
2068	USA	M	70	E	1
2072	USA	M	80	E	1

Then create a matrix as shown below to calculate this % for each combination of gender and category.

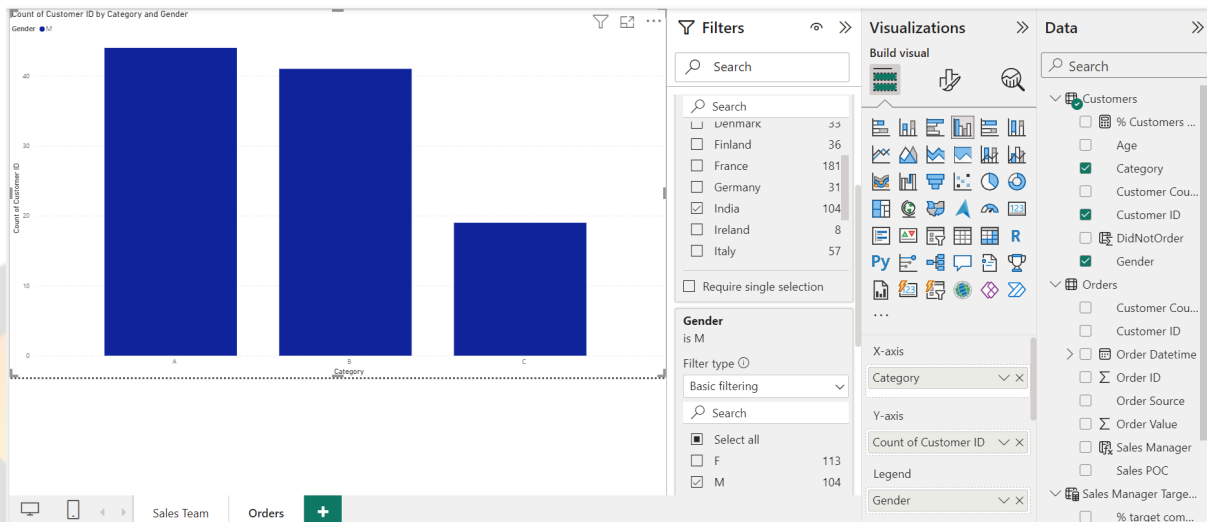
Category	F	M	Total
A	0.24	0.31	0.27
B	0.24	0.29	0.27
C	0.30	0.30	0.30
D	0.24	0.26	0.25
E	1.00	1.00	1.00
Total	0.40	0.44	0.42

10. Plot number of orders in each country with legend as Order Source.

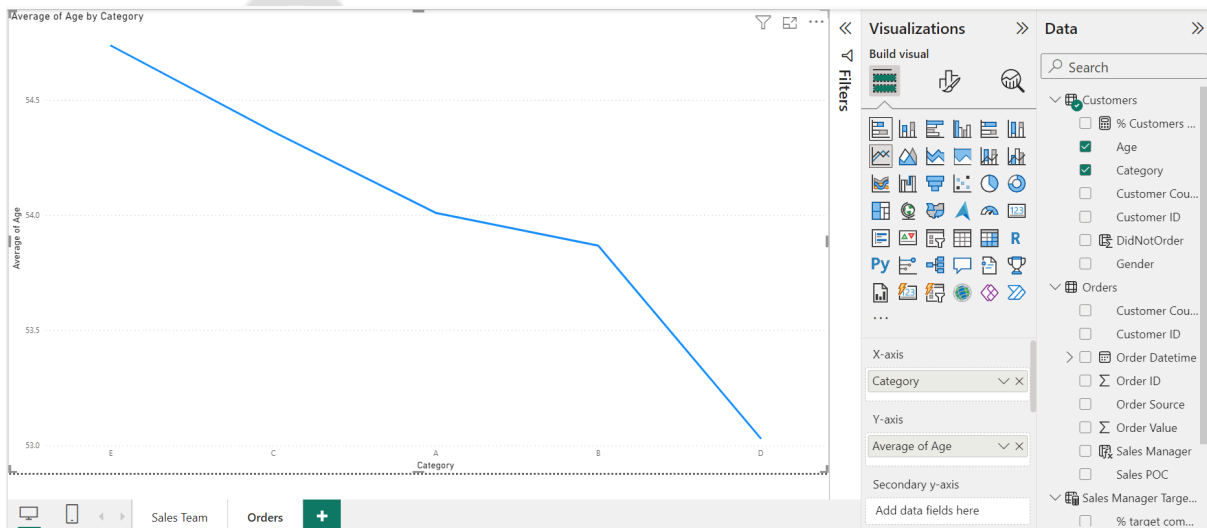


11.

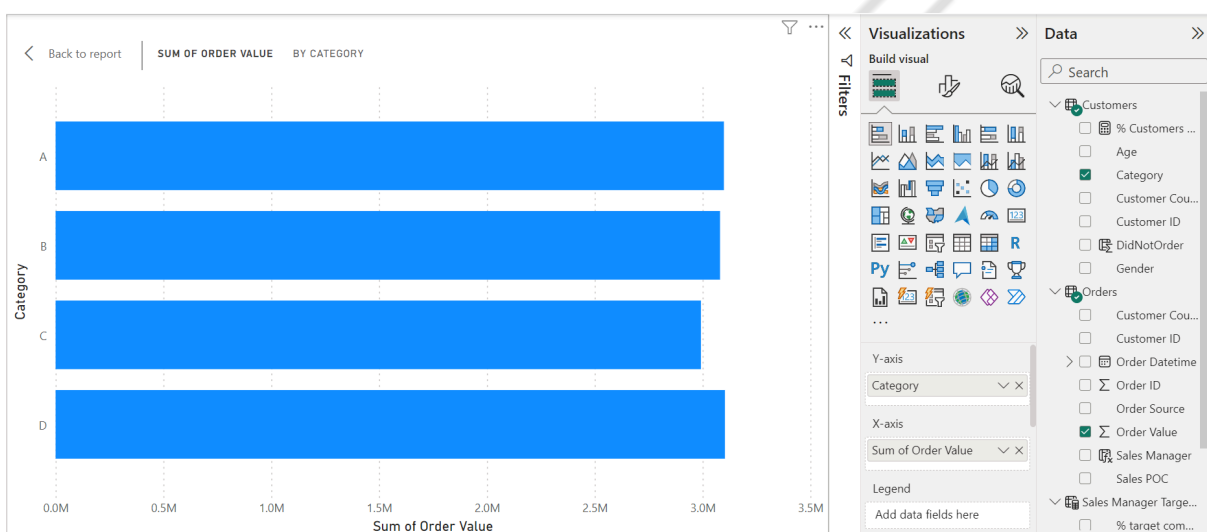
a. Plot count of customer ID by category with appropriate filters



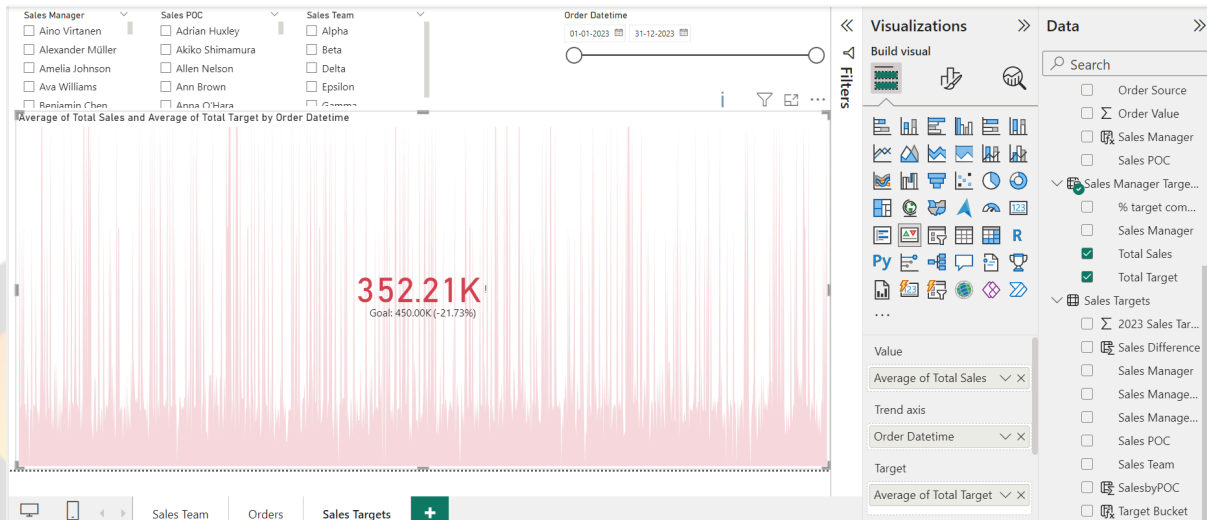
b. Find average age per category of customers:



c. Create a total order value by category chart and sort the category in Ascending order.



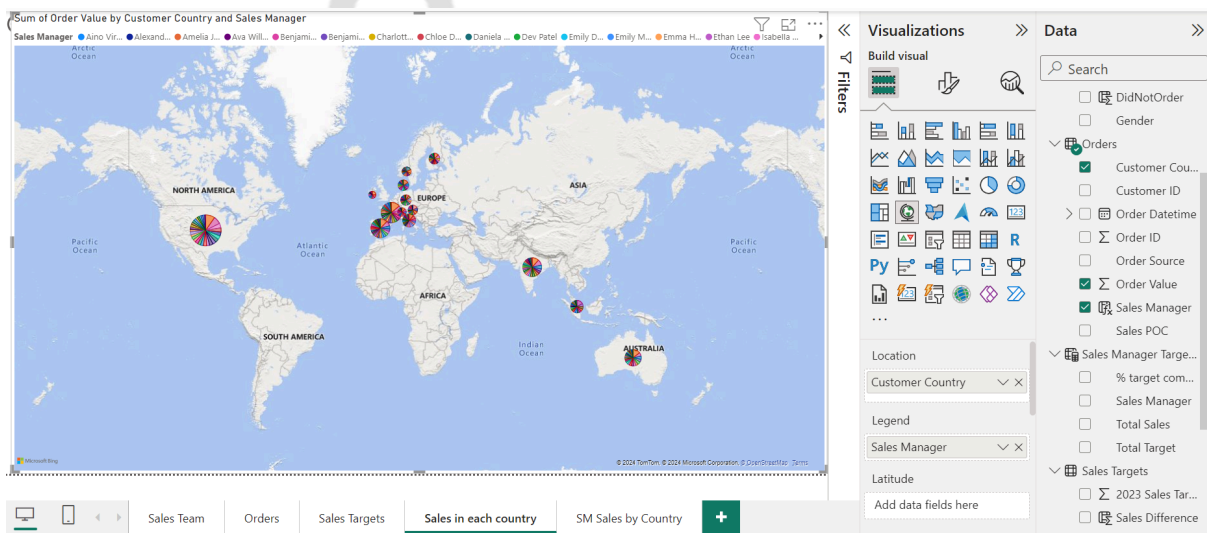
d.



12. Add a Q&A Visual and customize the questions if required.

13.

- a. Create a page with a map for total order value by country. The legend here must be the Sales Manager.

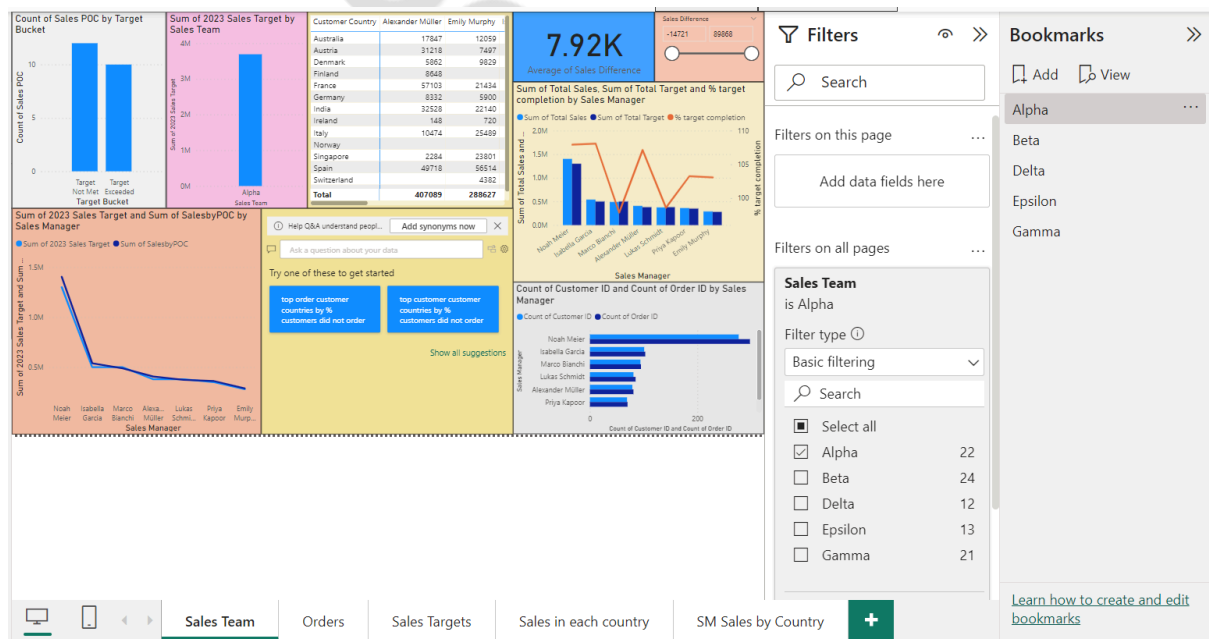


- b. Create another page for the drillthrough. On this page create a Matrix for total sales for each SM in each country. In the drillthrough fields, the Sales Manager must be added.

Sales Manager	Australia	Austria	Denmark	Finland	France	Germany	India	Ireland	Italy	Norway	Singapore	Spain	Switzerland	USA	Total
Aino Virtanen	25568		20833	6952	19104	5783	15484	6692	30360	24253	34689	14976	77627	77627	282411
Alexander Müller	17847	31218	5862	8648	57103	8332	32528	148	10474	2284	49718		162927		407089
Amelia Johnson	51461	1699	775		77104	11600	31549	7938	24085	8713	6791	43602	118427		383744
Ava Williams	16884	5053	19774	13074	63040	4492	60821		3349	13335	29859	8257	117947		355885
Benjamin Chen	71967	17625	20440	35714	122626	20791	133768	8148	32918	11732	83304	117924	17606	338017	1032580
Benjamin Weber	55946	642	17881	11080	63176	15669	62890	930	3481	1716	4650	60775	7486	153881	466003
Charlotte Miller	20220	3650	16993	23852	47750	6246	37348		22747	5886	18816	24002	2713	96997	327240
Chloe Dupont	6733	3924	10287	14395	44219	2664	58440		14908			30318		191805	377693
Daniela Hernandez	33641	3121		8408	30327	927	45270	1072	8522	35468	25965	36213	2228	159541	390703
Dev Patel	12480	15001	19499	7481	68188	13187	42277	5671	19166	19489	42084		135282		399805
Emily Dubois	23706	6399	19388	12685	36044	8615	49746		7637	3392	9161	54629	5162	86069	322633
Emily Murphy	12059	7407	9620		21434	5900	22140	720	25489		23801	56514	4382	98862	286527
Emma Hansen	1327	5113	6117	12261	60685	21843	45668		6578	11685		54455	9636	123066	358434
Ethan Lee	38465	6510	4953	8190	12523		40877		8180	6186	38844	43255	9873	142657	360513
Isabella Garcia	45277	6030		21308	84002	17355	46521	6063	23288	4019	15070	63903	8024	199550	540410
Liam Baumann	8514	6527		10333	33400	21961	5512		24856		8355	30383	576	122603	273020
Liam O'Sullivan	13972	9460	3774	10726	67991	2923	56097		13578	14242	13000	28446	6590	111407	352206
Lucas Nielsen	24557	37508	6046		50997	11259	32208		26537	10730		32398	8432	126321	366993
Lucas Olsen	36736			6807	14316	75	32330		18081		16657	30075	1369	163233	319679
Lukas Schmidt	54152		10933	799	54750	15706	35784		7879	4570	5523	59635		124873	374604
Marco Bianchi	58957	9085	21758	4854	34339	7173	88555	7588	18000	9467	13655	71134	8390	136112	489267
Mia Khan	10611	849	8983		9105		1641		11308			26075		85860	154432
Noah Meier	140715	26281	52893	42268	229185	5435	159140	23280	63721	26102	29838	169966	21489	412628	1402941
Oliver Kumar	42571		12867	70265		29338	3844	7311	13821	5782	37663	6696	52723		283001
Olivia Jensen	30439	5634	18559	1562	75264	8536	27671	3233	8592	5714	26353	31549	7479	131215	381800
Priya Kapoor	23812	856		17856	52413	11184	26367	4576	19320	7817	15490	24963	3073	153635	361362
Sofia Lane	29165			12781	23326	52316	18467	8118			32316	26275	237	104169	307888
Sophia Rossi	39856	7313	4046	17686	35680	26113	17619	12063	44369		20504	31763	11210	137471	406493
Sophie Gruber	24504		4370	9537	40711	19616	20920		5853	6296	11257		101332		244396
William Brown	2075	6921	2467	4856	51529	2725	54623		16563	6601	27277		88401		264038
Total	974217	223916	307260	324319	1640341	299656	1365448	91966	539764	195229	486633	1386799	165904	4274438	12275890

14. Check Solution PBIX file

15. Use Filters on all pages option to create 5 different views: one for each team and create a corresponding bookmark for each view



16. Mobile View can be created using the mobile icon at the bottom. Remember to keep visuals large and less in number on each page in the mobile view

